MIL-R-30500B

31 July 1972
SUPERSEDING
MIL-R-30500A
5 May 1966

MILITARY SPECIFICATION

ROPE, POLYESTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers polyester rope for general purpose use.

1.2 Classification.- The polyester rope shall be furnished in one type in the circumferences (size) specified (see 6.2).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock

MILITARY

MIL-C-3131 - Cordage; Preparation for Delivery of

STANDARDS

FEDERAL

FED-STD-191 - Textile, Test Methods

FSC 4020

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.1.1 and 6.4).

* 3.2 Government or supplier purchases.- The requirements specified in 3.8, 3.9 and 3.10 apply only to rope purchased directly by the Government. All other requirements apply to rope purchased as a component for an end item by a supplier and to rope purchased directly by the Government.

3.3 Standard sample.- Dyed rope shall match the standard for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.4 Materials.- The material used in the rope shall be bright, virgin, continuous filament polyester fiber of tenacity and denier to meet the requirements of this specification.

* 3.5 Physical requirements.- The rope shall be of three strands and shall conform to the requirements specified in table I when tested as specified in 4.2.5. Each strand shall be made of one size of one single polyester yarns in sizes 5/8 inch to 1 inch circumference inclusive and one size of balanced 3 ply polyester yarns in sizes from 1-1/8 inch circumference and up and the lay of the rope shall be right hand or "Z" lay.

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TABLE I.- Physical requirements

Circumference	Circumference tolerance of plus or minus	Diameter 1/ (approx.)	Length per pound 2/ (min.)	Breaking strength (min.)	Length per reel (min.)	Weight per reel (approx.) 1/ 3/	Hardness (min.)	Hardness (max.)
Inches	Inches	Inches	Feet	Pounds	Feet	Pounds	Pounds	Pounds
5/8	1/16	3/16	84.0	800	2,250	28	5	25
3/4	1/16	1/4	50.0	1,200	2,250	48	5	25
1	1/16	5/16	33.0	2,500	2,250	72	5	25
1-1/8	1/16	3/8	23.5	2,800	1,600	73	5	25
1-1/4	1/16	7/16	17.0	3,800	1,200	74	5	25
1-1/2	1/16	1/2	13.0	5,000	1,200	97	5	25
1-3/4	1/16	9/16	10.0	6,500	1,200	126	5	25
2	1/8	5/8	8.0	8,000	1,200	158	5	25
2-1/4	1/8	3/4	6.0	10,000	1,200	210	5	25
2-1/2	1/8	13/16	5.3	13,000	1,200	238	5	25
2-3/4	1/8	7/8	4.2	15,000	1,200	300	5	25
3	1/8	1	3.5	18,500	1,200	360	20	100
3-1/2	3/16	1-1/8	2.5	25,000	1,200	504	20	100
4	3/16	1-5/16	2.0	31,000	600	315	20	100
5	1/4	1-5/8	1.30	48,000	600	485	20	100
6	5/16	2	0.85	68,000	600	742	20	100
7	3/8	2-1/4	0.66	88,000	600	956	20	100
8	3/8	2-5/8	0.50	110,000	600	1,260	20	100
9	7/16	3	0.40	140,000	600	1,575	20	100
10	1/2	3-1/4	0.33	165,000	600	1,910	20	100
11	9/16	3-1/2	0.25	240,000	600	2,400	20	100
12	5/8	4	0.20	285,000	600	2,800	20	100

1/ Not a specification requirement - included for information purposes only.
2/ A minus tolerance of 10 percent is allowed for dyed rope.
3/ A plus tolerance of 10 percent shall be allowed for dyed rope.

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* 3.5.1 Elongation.- The elongation of the rope shall not exceed 35 percent when tested as specified in 4.2.5.

3.5.2 Hardness.- The rope shall meet the minimum and shall not exceed the maximum hardness requirements specified in table I, when tested as specified in 4.2.5.

3.6 Finish.- No extraneous material shall be added for the purpose of weighting the rope. The extractable matter of the finished rope shall not exceed 4.0 percent when tested as specified in 4.2.5.

3.7 Color.- Unless otherwise specified (see 6.2), the color of the finished rope shall be natural. When colored rope is specified the rope shall be dyed to match an approved standard shade of Olive Drab No. 7 (see 6.3).

3.7.1 Colorfastness.- Dyed and finished rope shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.2.5. When no standard sample has been established, dyed rope shall show good fastness to weathering when tested as specified in 4.2.5.

* 3.8 Identification.- Each reel (spool) of rope shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15-point paper stock and have a minimum tearing resistance of both directions (total) of 850 grams when tested as specified in UU-T-81. The label shall be attached in such a manner as to remain in place and be clearly legible until all rope has been removed. All ticket or label entries shall be legibly printed, stamped or typed with water insoluble ink. The ticket or label shall contain the following information:

Stock number
Item description
Specification number
Length and circumference (size)
Contract number and date
Date of manufacture (Month and Year)
Supplier's name

3.8.1 Identification marker.- The manufacturer shall identify his product by inserting a kraft paper or water repellent cotton marker within one strand in all ropes larger than 1-1/8 inch circumference. The marker shall be completely enveloped by the cover yarns in the strand in which it is inclosed. Unless otherwise specified, the manufacturer's name, year of manufacture and type of fiber (POLYESTER) shall be clearly printed in block type on the marker. The printing shall not be affected upon exposure to water or mineral oil.

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3.9 Put-up.- Unless otherwise specified, (see 6.2) the rope shall be furnished without knots or splices on non-returnable reels (spools) not larger than 6 feet in diameter in the lengths specified in table I. Pieced lengths will be permitted, but no piece shall be less than 600 feet when measured in the relaxed condition. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The reels (spools) shall be wound so that each turn and layer is free from entanglement.

3.9.1 When specified, rope shall be put up in coils in one continuous piece without knots or splices in the minimum length specified (see 6.2). A plus tolerance of 10 percent shall be allowed on the length per coil. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The coils shall be so wound that each turn and layer is free from entanglement.

3.10 Moisture content.- The moisture content of the rope, as received, shall not exceed 2.0 percent when tested as specified in 4.2.5 (see 6.5).

3.11 Spliceability.- The finished rope shall be spliceable and shall not develop yarn displacement or strand cockles in the splicing test specified in 4.2.5.

3.12 Workmanship.- The rope shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels established by this specification.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 First article inspection.- When required (see 6.2), the preproduction sample submitted in accordance with 3.1 shall be visually inspected for appearance and color. The sample shall be tested for physical and chemical characteristics in accordance with 4.2.5.

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* 4.1.2 Certificate of compliance.- When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in table II and a supplier's certificate of compliance thereto shall be furnished and will be accepted for the requirements.

TABLE II.- Component test

Characteristic	Paragraph reference
Material	3.4
Yarn size	3.5
Single yarn in strand (when applicable)	3.5
Balanced plied yarn in strand (when applicable)	3.5

* 4.2.2 Examination of the end item for visual defects.- The defects listed in table III shall be counted regardless of their proximity to each other. The sample unit shall be one reel (spool) or coil as applicable. Ten percent of the length contained on each sample unit, but not less than 100 feet shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of reels (spools) or coils each. The acceptable quality level (AQL) shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

TABLE III.- Visual examination defects

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Cut, any	X	
	Chafed or damaged	X	
	Kink, darting yarn 1/, broken or loose end, bulged strand, strand knot	X	

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TABLE III.- Visual examination defects (cont'd)

Examine	Defect	Classification Major	Minor
Appearance and workmanship (cont'd)	Ends not cut off squarely		X
	Ends not securely whipped, taped or heat sealed		X
	Splice or otherwise secured to make a continuous length	X	
Identification	Omitted, incorrect, or illegible		X
	Insecurely attached or not as specified		X
Color	Not as specified	X	
	Resists and/or spotty dyeing		X
	Specific shade not within established tolerances		X
	Uneven		X
Cleanness	Spot, or stain, clearly noticeable 2/		X

1/ Darting yarn is an internal yarn which projects through the cover of the strand.

2/ At normal inspection distance (approximately 3 feet).

4.2.3 Examination for length and winding.- The sample unit for this examination shall be one reel (spool) or coil. The inspection level shall be level S-3 and the acceptable quality level shall be 4.0 percent defective. The minimum sample size shall be code letter "E" (sample size 13, acceptance number 1). The lot size shall be the number of units in the inspection lot. Defects shall be as listed in 4.2.3.1 and 4.2.3.2. The required length of the spool (reel) or coil (see table I and 3.9.1) shall be determined using the following formula:

$W \times L$ = Length per holder in feet

Where: W is net weight of the rope in 0.1 pounds

L is the lot average of the weight-length relationship (length/pound) as determined in 4.2.5

4.2.3.1 Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

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a. Length of spool (reel) less than specified.

b. Length of coil less than or more than length specified (including permitted tolerances).

c. Length of unit less than marked on ticket.

d. Rope not in continuous length when specified (see 3.9.1).

e. Any piece on reels (spools) less than 600 feet in length.

4.2.3.2 Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

a. Improperly or not firmly wound resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of rope.

b. Knot or splice or otherwise joining of ends to make a continuous length.

* 4.2.4 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging, packing and marking comply with the section 5 requirements of this specification. The examination shall be in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 defects per 100 units.

* 4.2.5 Testing of the end item.- The methods of testing specified in FED-STD-191, wherever applicable and as listed in table V shall be followed. When data in the "Number of determinations" and "Results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table IV. The sample unit for testing shall be one reel (spool) or coil of rope. The lot size shall be expressed in units of one reel (spool) or coil. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In cases of dispute, tests shall be made upon material which has reached equilibrium under Standard Conditions as defined in FED-STD-191.

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TABLE IV.- Sampling for tests

Number of Reels (spools) or coils in lot	Number of samples
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE V.- Test methods

Characteristic	Requirement reference	Test method
Number of strands	3.5	Visual 1/
Lay of rope	3.5	Visual 1/
Circumference	3.5	6003
Length per pound	3.5	6004
Breaking strength	3.5	6015 2/
Elongation	3.5.1	6015 2/
Hardness	3.5.2	6020
Extractable matter	3.6	2611 3/
Color (shade) matching	3.7	4.2.5.1 1/
Colorfastness to weathering	3.7.1	5671 4/
Identification marker	3.8.1	Visual 5/
Moisture content	3.10	2600
Spliceability	3.11	4.2.5.2 1/

1/ One determination per sample unit shall be made and the results shall be reported as pass or fail.

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2/ Method 6016 of FED-STD-191 may be used to determine breaking strength and elongation for ropes 3 inches in circumference or less.

3/ Except that extractable matter shall be determined using the chloroform extraction procedure.

4/ The time of exposure shall be 40 standard fading hours.

5/ Supplier's certificate of compliance thereto shall be furnished and will be accepted for the requirement.

* 4.2.5.1 Determination of shade (color) matching.- The shade of the dyed rope shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be good approximation to the standard sample under incandescent lamplight at 2800° Kelvin (see 6.3).

4.2.5.2 Determination of spliceability.- A breaking strength specimen shall be prepared and spliced at each end with four full tucks. The specimen shall then be allowed to rest for 24 hours. Three of the tucks shall then be backed out of the rope. That portion of the rope from which the tucks are removed shall be examined for yarn displacement and strand cockles.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The rope, put-up as specified, shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- The rope shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking.- In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

6. NOTES

6.1 Intended use.- Rope covered by this specification is intended for general purpose use.

6.2 Ordering data.- Rope covered by this specification is intended for general purpose use.

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(a) Title, number and date of this specification.
(b) Circumference (size) required (see table I).
(c) First article, when first article sample is required (see 3.1, 4.1.1, and 6.4).
(d) Color required, if other than natural (see 3.7).
(e) Put-up and length when other than specified (see 3.9).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(g) That polyester rope shall be purchased on a price-per-pound basis net weight.
(h) That purchaser will accept at original weight, any unit which has been shortened or cut for test specimens, if in compliance with this specification.

6.3 Standard sample.- For access to the standard sample, address the procuring office issuing the invitation for bids.

6.4 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one spool (reel) or coil of rope. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Adjustment for high moisture content.- Material furnished containing an excess of moisture will be accepted by an adjustment in weight to the 2 percent moisture basis.

6.6 Marginal notations.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SH
Air Force - 82

Preparing activity:

Army - GL

Project No. 4020-0210

Review activities:

Army - WC
DSA - IS

User activity:

Army - ME

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Rope, Polyester MIL-R-30500E

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity - Optional)	DATE

DD FORM 1426

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